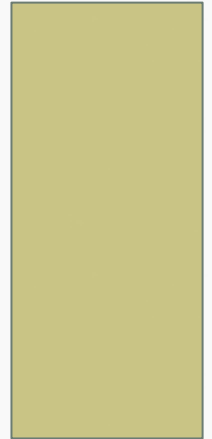


CHAPTER 1 - INTRODUCTION

INFO 4401

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1.0 PREAMBLE

“An introduction or preface..”

How to write your preamble?

“States the reasons for and intent of the document.”

“Provide an overview of your report”

“Define the topic/key terms”

“Explain the value of your study”



1.0 PREAMBLE

Defining topic/key term

Project name:iHeart

iHeart is a heart rate monitoring mobile application that utilizes a heart rate band. It is specifically designed and developed for the **use of elderly people** who does not require complex features and interface, but rather **significant features that may help them in daily monitoring and emergency situation.**

1.1 PROBLEM DESCRIPTION

- **1.1 PROBLEM DESCRIPTION**
 - 1.1.1 Background of the problem
 - 1.1.2 Problem statement

“Clear description of the issue..”

“Use 5Ws (What? When? Why? Who? Where?) to spark discussions..”

- **Who will be affected?**
- **Why is it important to be fixed**
- **What will be the outcome if it is not fixed**
- **Where the problem is taking place**
- **When the problem needs to be fixed**

Problematic Problem Statement

Problem statement of an event management website

“Based on the developers experience as student of IIUM, the student need through **many procedure** to make an event in university. Student need to fill **many forms to get approval from the department** to booking the place”



Examples of Good Problem Statement

Project name: myCare (edited)

1.1.1. BACKGROUND OF THE PROBLEM

Patients of upscale or boutique medical centers, such as Gleneagles or Prince Court Kuala Lumpur, expect the best in service when they chose those institutions. However, patients who do not enroll as an in-patient do not get the same care they would receive as an outpatient client. Out-patient clients are responsible for the tracking of their medication or recovery. As such, there needs to be an app for the healthcare industry.

1.1.2. PROBLEM STATEMENT

Currently there is **no existing app** launched in Malaysia with these specific objectives. As a result, patients are left to track their care process manually. This may result in **incorrect dosage** of medication, **missed appointments** with their care team, as well as a **lack of data** or **progress for them to track** after they have left the facilities.

1.2 PROJECT OBJECTIVES

Project objectives must be SMART!

S - Specific (What do you want to accomplish?)

M - Measurable (How can you measure progress?)

A - Achievable (Do you have the skills required?)

R - Relevant (Why am I setting this goal right now?)

T - Time-bound (What's the deadline? Is it realistic?)

Examples

Project name: myCare

1.2 Project objectives

By the end of this project, the developer should be able to at least provide a prototype of an app that may work with existing medical data systems to help track the medication and care of recovering patients. Reports to be produced include surveys, as well as usability testing. The app should be able to automate the process of tracking the consumption of medication, tracking the occurrence of medical symptoms, provide insights on the recovery progress of the patient, as well as communication between patients and their Care Teams.

1.3 PROJECT SCOPE

- **1.3 PROJECT SCOPE**
 - **1.3.1 Scope (your coverage)**
 - **1.3.2 Target Audience (define who)**
 - **1.3.3 Specific Platform (define the tools)**

Examples: Scope

Project name: iHeart

iHeart parent view

a. parent will be able to monitor heart rate, push on an emergency button for direct call to their caregiver, sent text messages and notification on heart rate alarm.

iHeart caregiver view

Caregiver will be able to view user heart rate reading, user's weekly summary report of his/her heart rate reading, sent text messages and direct call to the user.

Examples: Scope

Project name: Dzikra Faith

Dzikra Faith User View

- a. User will be able to use the Android application to track and remind them of their prayers time
- b. User will be able to get summaries of their prayer daily, weekly, monthly

Dzikra Faith Admin View

Admin will be able to login, update version, revise comment/feedback and post advertisement

Examples: Target audience

Project name: iMaalum

All undergraduate and postgraduate students of IIUM.

Project name: iHeart (edited)

Targeted users are:-

- 1) elderly people (50-70 years old), especially those with heart diseases; and
- 2) their caregivers.

Examples: Specific platform

Be generic....

Project name: iMaalum

The application will be developed using React Native. Thus, both Android and iOS devices can run the application. It is recommended for the user to have either Android version 4.1 or higher, or iOS version 9 or higher.

Specific platform

Or specific....

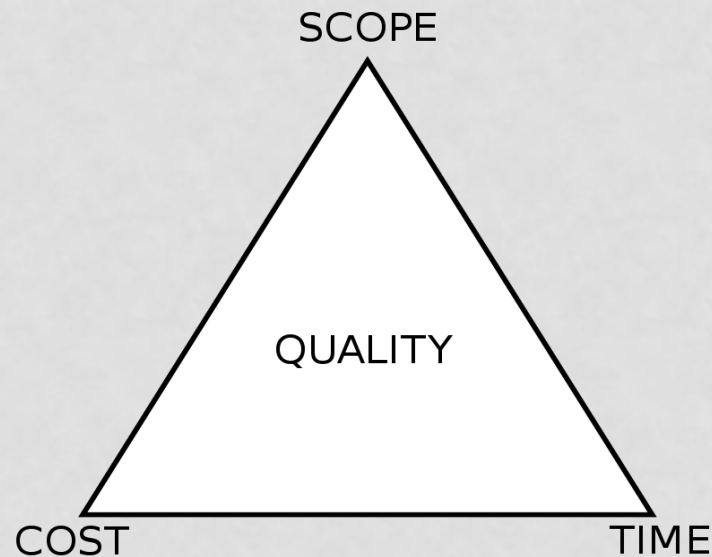
Project name: Caesar

Software	Description
Microsoft Windows 8	Operating system environment
Adobe Photoshop CS 6	Editing the interface
Microsoft Word 2014	Writing report
Microsoft Project	Creating Gantt chart
Microsoft Powerpoint	Presentations
StarUML	Creating usecase diagrams
ApplInventor	Prototyping the application
Creately	Creating diagrams

Hardware? Processor, memory, sound/graphic cards

1.4 Constraints

THREE project constraints:-



or...

SIX project constraints:-

Time

Scope

Cost

Risk

Quality

Resources

But I am not supposed to mention TIME as a constraint!!!

Time as a constraint??

Project name: EEG system for fitness and health (edited)

“EEG system is new knowledge for the developers and so developers are **required to learn and understand the general about EEG.**

Moreover, **developers also need to learn and practice python language in very short period.** These constraints may affect timely completion of project tasks.”

Project name: myCare (edited)

“A major constraint includes **knowledge of app development for Apple mobile devices, as currently only a rough prototype exists.** Consequently, time may also be a constraint.”

And more constraints...

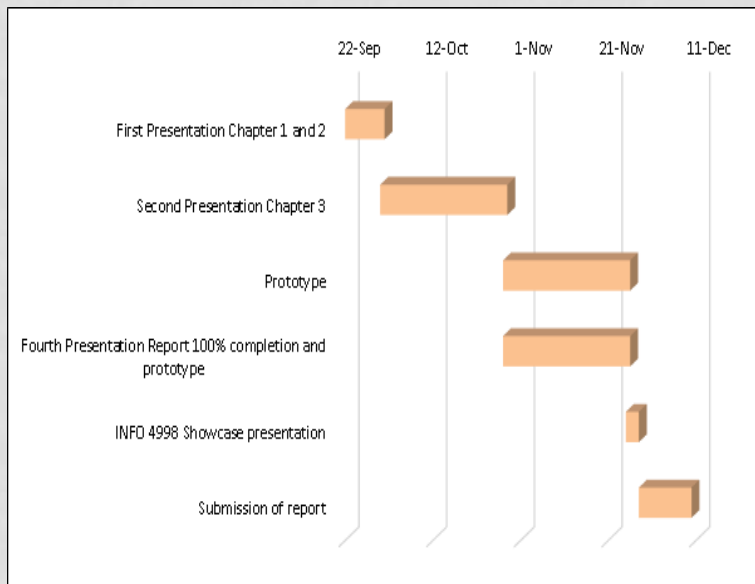
Project name: Class schedule and study plan generator

In the development process that includes project requirement gathering and project specifications, the collaboration with ITD department of IIUM restrict the progress flow. The strict security of IIUM database causing a hardship in developing database based system. Not enough information on metadata and few data that is considered as private by the university limit the development team to work on the output.

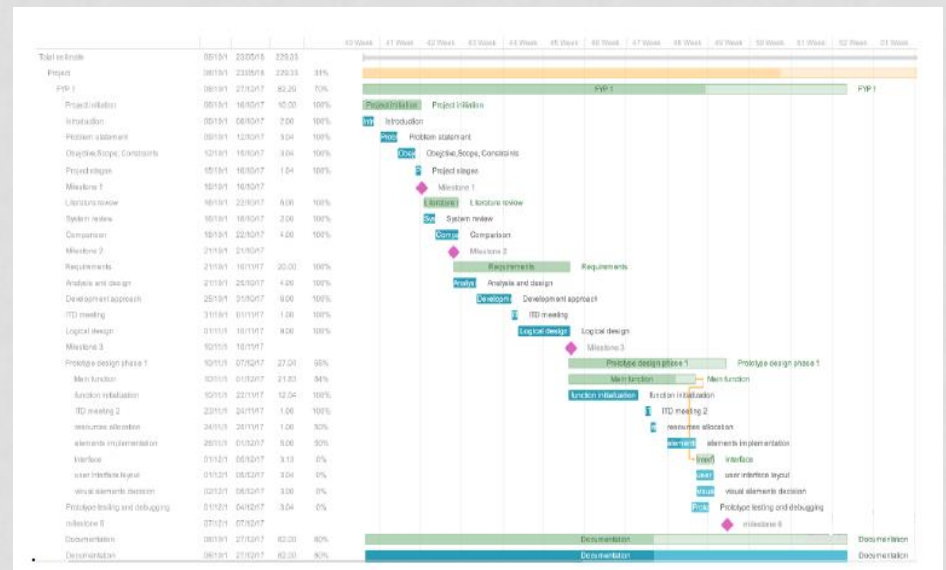
1.5 Project Stages

The timeline of your project.

Can be accompanied with illustration/diagram (choose ONE diagram only).



Source: Short Form Checker



Source: Caesar

1.6 Significance of the project

There are many ways to write project significance. Most students appear to write the following:-

- A) increase understanding or knowledge in a certain area;
- B) improve organizations' services ;
- C) contribution to organization policy;
- D) explain benefits of the project; or
- E) who likely benefits from the project

Significance??

Project significance of a mobile application:-

The significance of this system are as follows:-

- Easy to access
- Benefits in many ways
- Provide knowledge to the users
- Users can also share the knowledge to others

1.6 Significance of the project

Project name: EEG system for health and fitness (edited)

“User friendly display of health information that allow users to understand the link between their emotions and fitness activities”

Project name: myCare (edited)

“..patients using the app can automatically share their progress and physicians can provide advice in return; thereby increasing personal relationship between patients and their assigned Care Team..”

Project name: Class schedule and study plan generator

“..Student can make a better decision in choosing the right courses for every semester”

1.7 Project summary

Project name: myCare (edited)

(Problem)

There is no efficient way for patients to track the process of their medication or recovery. In fact, this causes problems such as ...

(Objectives)

myCare when launched should be able to interface with ...

(Significance)

This app should provide a three-fold benefit;...

(Scope)

myCare is built using Xcode, targeting iPhone 5s and ...

(Constraint)

Constraints include the skill or knowledge required, ...

Acknowledgements

- Dr Noor Hayani; for the FYP policy and Chapter 1 template ;
- My FYP students, especially projects quoted in this presentation:
 - myCare
 - iHealth
 - Dzikra Faith
 - Caesar
 - iMaalum
 - EEG system for health and fitness
 - Class schedule and study plan generator
 - Short Form Checker

Demo

- Reference Manager using Mendeley – APA6th Edition.
- Microsoft word for reporting
- Project Scheduling: Gantt Chart
- FYP Student Dashboard

Tips

- My own experience:
 - 1) Know what we are doing.
 - 2) Know what you suppose to deliver
 - 3) Plan and re-plan
 - 4) Keep writing!...keep develop!.. baby steps will do
 - 5) Meeting & Cooperation (supervisor, teammate, peers)
 - 6) Don't be alone...Seek help!
 - 7) Deliver!....Don't delay
 - 8) Have fun!!

Questions?

